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63542028
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MariaDB [(none)]> create database Projects;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> use Projects;
Database changed
MariaDB [Projects]> create table Department(deptNo VARCHAR(10) PRIMARY KEY,
deptName VARCHAR(50));
Query OK, 0 rows affected (0.019 sec)

MariaDB [Projects]> desc Employee;

Field	Type	Null	Key	Default	Extra
empNo	varchar(10)	NO	PRI	NULL	
firstName	varchar(30)	YES		NULL	
lastName	varchar(30)	YES		NULL	
Address	varchar(60)	YES		NULL	
birthday	date	YES		NULL	
sex	varchar(1)	YES		NULL	
position	varchar(25)	YES		NULL	
deptNo	varchar(10)	YES	MUL	NULL	
salary	decimal(10,2)	YES		NULL	
retirementDate	date	YES		NULL	

10 rows in set (0.036 sec)

MariaDB [Projects]> CREATE TABLE Project(projNo VARCHAR(10) PRIMARY KEY, projName
VARCHAR(50), deptNo VARCHAR(10), FOREIGN KEY (deptNo) REFERENCES
Department(deptNo));
Query OK, 0 rows affected (0.017 sec)

MariaDB [Projects]> desc Project;

Field	Type	Null	Key	Default	Extra
projNo	varchar(10)	NO	PRI	NULL	
projName	varchar(50)	YES		NULL	
deptNo	varchar(10)	YES	MUL	NULL	

3 rows in set (0.032 sec)

MariaDB [Projects]> CREATE TABLE WorksOn(empNo VARCHAR(10), projNo VARCHAR(10),
dateWorked DATE, hoursWorked INT CHECK (hoursWorked >= 0 AND hoursWorked <= 40),
PRIMARY KEY(empNo, projNo, dateWorked), FOREIGN KEY(empNo) REFERENCES

```
Employee(empNo), FOREIGN KEY(projNo) REFERENCES Project(projNo));
Query OK, 0 rows affected (0.023 sec)
```

```
MariaDB [Projects]> desc WorksOn;
```

Field	Type	Null	Key	Default	Extra
empNo	varchar(10)	NO	PRI	NULL	
projNo	varchar(10)	NO	PRI	NULL	
dateWorked	date	NO	PRI	NULL	
hoursWorked	int(11)	YES		NULL	

```
4 rows in set (0.035 sec)
```

```
MariaDB [Projects]> INSERT INTO Department VALUES ('D1', 'IT'), ('D2', 'Planning'),
('D3', 'Finance'), ('D4', 'Marketing');
Query OK, 4 rows affected (0.005 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
MariaDB [Projects]> select * from Department;
```

deptNo	deptName
D1	IT
D2	Planning
D3	Finance
D4	Marketing

```
4 rows in set (0.001 sec)
```

```
MariaDB [Projects]> INSERT INTO Employee VALUES
```

```
-> ('E1', 'Kwame', 'Mensah', '123 Independence Ave, Accra', '1985-05-15', 'M',
'Manager', 'D1', 2500.00, '2025-12-31'),
-> ('E2', 'Akosua', 'Boateng', '456 Liberation Rd, Kumasi', '1992-08-20', 'F',
'Team Leader', 'D1', 1800.00, '2040-06-15'),
-> ('E3', 'John', 'Smith', '789 Freedom St, Takoradi', '1978-03-10', 'M',
'Analyst', 'D2', 1200.00, '2035-09-01'),
-> ('E4', 'Abena', 'Kusi', '321 Unity Ave, Cape Coast', '1995-11-25', 'F',
'Software Developer', 'D1', 1600.00, '2045-04-20'),
-> ('E5', 'David', 'Brown', '654 Progress Dr, Tamale', '1980-07-08', 'M',
'Manager', 'D3', 3000.00, '2025-07-01'),
-> ('E6', 'Efua', 'Ansah', '987 Peace Ln, Ho', '1990-01-12', 'F', 'Analyst',
'D4', 1100.00, '2050-03-15'),
-> ('E7', 'Michael', 'Davis', '147 Hope Blvd, Koforidua', '1988-09-30', 'M',
'Team Leader', 'D1', 1700.00, '2042-11-30'),
-> ('E8', 'Akua', 'Frimpong', '258 Wisdom Way, Bolgatanga', '1975-04-18', 'F',
'Manager', 'D2', 2700.00, '2025-10-15'),
-> ('E9', 'Nana', 'Gyasi', '369 Strength St, Wa', '1993-07-22', 'M', 'Software
Developer', 'D3', 1900.00, '2055-03-20'),
```

```

-> ('E10', 'Sarah', 'Jones', '753 Courage Rd, Sunyani', '1991-12-14', 'F',
'Analyst', 'D4', 1300.00, '2052-08-10');
Query OK, 10 rows affected (0.011 sec)
Records: 10 Duplicates: 0 Warnings: 0

```

```

MariaDB [Projects]> select * from Employee;

```

```

+-----+-----+-----+-----+-----+-----+-----+
| empNo | firstName | lastName | Address | birthday | sex |
| position | deptNo | salary | retirementDate |
+-----+-----+-----+-----+-----+-----+-----+
| E1 | Kwame | Mensah | 123 Independence Ave, Accra | 1985-05-15 | M |
Manager | D1 | 2500.00 | 2025-12-31 |
| E10 | Sarah | Jones | 753 Courage Rd, Sunyani | 1991-12-14 | F |
Analyst | D4 | 1300.00 | 2052-08-10 |
| E2 | Akosua | Boateng | 456 Liberation Rd, Kumasi | 1992-08-20 | F |
Team Leader | D1 | 1800.00 | 2040-06-15 |
| E3 | John | Smith | 789 Freedom St, Takoradi | 1978-03-10 | M |
Analyst | D2 | 1200.00 | 2035-09-01 |
| E4 | Abena | Kusi | 321 Unity Ave, Cape Coast | 1995-11-25 | F |
Software Developer | D1 | 1600.00 | 2045-04-20 |
| E5 | David | Brown | 654 Progress Dr, Tamale | 1980-07-08 | M |
Manager | D3 | 3000.00 | 2025-07-01 |
| E6 | Efua | Ansah | 987 Peace Ln, Ho | 1990-01-12 | F |
Analyst | D4 | 1100.00 | 2050-03-15 |
| E7 | Michael | Davis | 147 Hope Blvd, Koforidua | 1988-09-30 | M |
Team Leader | D1 | 1700.00 | 2042-11-30 |
| E8 | Akua | Frimpong | 258 Wisdom Way, Bolgatanga | 1975-04-18 | F |
Manager | D2 | 2700.00 | 2025-10-15 |
| E9 | Nana | Gyasi | 369 Strength St, Wa | 1993-07-22 | M |
Software Developer | D3 | 1900.00 | 2055-03-20 |
+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.001 sec)

```

```

MariaDB [Projects]> INSERT INTO Project VALUES

```

```

-> ('P1', 'SCCS', 'D1'),
-> ('P2', 'E-Commerce Platform', 'D1'),
-> ('P3', 'HR Management System', 'D2'),
-> ('P4', 'Financial Analysis Tool', 'D3'),
-> ('P5', 'Digital Marketing Campaign', 'D4');

```

```

Query OK, 5 rows affected (0.004 sec)
Records: 5 Duplicates: 0 Warnings: 0

```

```

MariaDB [Projects]> select *from Project;

```

```

+-----+-----+-----+
| projNo | projName | deptNo |
+-----+-----+-----+
| P1 | SCCS | D1 |

```

P2	E-Commerce Platform	D1
P3	HR Management System	D2
P4	Financial Analysis Tool	D3
P5	Digital Marketing Campaign	D4

5 rows in set (0.001 sec)

MariaDB [Projects]> INSERT INTO WorksOn VALUES

```
-> ('E1', 'P1', '2024-01-01', 35),
-> ('E2', 'P1', '2024-01-01', 30),
-> ('E4', 'P1', '2024-01-01', 25),
-> ('E1', 'P2', '2024-01-02', 20),
-> ('E2', 'P2', '2024-01-02', 18),
-> ('E4', 'P2', '2024-01-02', 15),
-> ('E7', 'P2', '2024-01-02', 22),
-> ('E3', 'P3', '2024-01-03', 28),
-> ('E8', 'P3', '2024-01-03', 32),
-> ('E5', 'P4', '2024-01-05', 38),
-> ('E9', 'P4', '2024-01-05', 25),
-> ('E6', 'P5', '2024-01-07', 22),
-> ('E10', 'P5', '2024-01-07', 18);
```

Query OK, 13 rows affected (0.012 sec)

Records: 13 Duplicates: 0 Warnings: 0

MariaDB [Projects]> select* from Workson;

empNo	projNo	dateWorked	hoursWorked
E1	P1	2024-01-01	35
E1	P2	2024-01-02	20
E10	P5	2024-01-07	18
E2	P1	2024-01-01	30
E2	P2	2024-01-02	18
E3	P3	2024-01-03	28
E4	P1	2024-01-01	25
E4	P2	2024-01-02	15
E5	P4	2024-01-05	38
E6	P5	2024-01-07	22
E7	P2	2024-01-02	22
E8	P3	2024-01-03	32
E9	P4	2024-01-05	25

13 rows in set (0.000 sec)

MariaDB [Projects]> CREATE VIEW FemaleManagedProjects AS SELECT p.* FROM Project p, Department d, Employee e WHERE p.deptNo = d.deptNo AND e.deptNo = d.deptNo AND e.sex = 'F' AND e.position = 'Manager' ORDER BY p.projNo;

Query OK, 0 rows affected (0.017 sec)

MariaDB [Projects]> CREATE VIEW EmployeeProjectDetails AS SELECT e.empNo,

```
e.firstName, e.lastName, p.projName, w.hoursWorked FROM Employee e, Project p,
WorksOn w WHERE e.empNo = w.empNo AND p.projNo = w.projNo;
Query OK, 0 rows affected (0.009 sec)
```

```
MariaDB [Projects]> CREATE VIEW EmpProject(empNo, projNo, totalHours) AS SELECT
w.empNo, w.projNo, SUM(hoursWorked) FROM Employee e, Project p, WorksOn w WHERE
e.empNo = w.empNo AND p.projNo = w.projNo GROUP BY w.empNo, w.projNo;
Query OK, 0 rows affected (0.017 sec)
```

```
MariaDB [Projects]> SELECT * FROM EmpProject;
```

empNo	projNo	totalHours
E1	P1	35
E1	P2	20
E10	P5	18
E2	P1	30
E2	P2	18
E3	P3	28
E4	P1	25
E4	P2	15
E5	P4	38
E6	P5	22
E7	P2	22
E8	P3	32
E9	P4	25

```
13 rows in set (0.012 sec)
```

```
MariaDB [Projects]> SELECT projNo FROM EmpProject WHERE projNo = 'SCCS';
Empty set (0.001 sec)
```

```
MariaDB [Projects]> SELECT COUNT(projNo) FROM EmpProject WHERE empNo = 'E1';
```

COUNT(projNo)
2

```
1 row in set (0.010 sec)
```

```
MariaDB [Projects]> SELECT empNo, totalHours FROM EmpProject GROUP BY empNo;
```

empNo	totalHours
E1	35
E10	18
E2	30
E3	28
E4	25

E5	38
E6	22
E7	22
E8	32
E9	25

+-----+

10 rows in set (0.001 sec)

7.28